H2S-NN: Heterogeneous Heterophilic Sheaf Neural Network with Path Instances

No edit

Progress report · target venue IJCAI-27

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Institute of SCIENCE TOKYO

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Reviewer feedback on the submitted version

R1
Weak Accept
main concern: scalability

R2
Weak Reject
motivation vs mechanism

R3
Weak Accept
heterophily evidence, cost

R4
Weak Reject
setting, isolation, cost

→

Meta-review
Reject
split reviews; key claims need support

Strengths the reviewers agreed on

Strength	Raised by
Path instance kept as a cell, not an endpoint relation: novel	R1–R4, MR
Principled construction, with a proof	R1, R2, R4
Three tasks, many datasets	R1–R4
Cost analysis given; code released	R2

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Reviewer feedback: the five main concerns

#	Concern (reviewers' wording, condensed)	Raised by
1	Heterophily not shown: only IMDB strongly heterophilic; no gain-vs-heterophily analysis	MR, R2, R3, R4
2	Face not isolated: it reaches nodes only through the closing (endpoint) edge	MR, R2, R4
3	No scalability study: ACM 9.3 GB, 3.0 s/epoch	MR, R1, R3, R4
4	Protocol: baseline tuning unclear; HetSheaf quoted; std unclear	MR, R1, R2, R4
5	Motivation: no failing example; extra modules not motivated	R2, R3, R4

Concerns 1–3 are about the core claim; 4–5 about how it was tested and explained.

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Update since last meeting

Introduction & contribution	Methodology	Path sampling	Experiments
before **3 contributions** H2S-NN + AMPSS + SheafCovPool	**before** **closing edge, K steps** face reached nodes via closing edge	**before** **AMPSS** schema search + 4 budgets	**before** **3 + 4 + 4 datasets** HetSheaf quoted; tuning unclear
↓	↓	↓	↓
now **1 core idea + WISP** paths as faces, original graph	**now** **one energy, one solve** matrix-free maps	**now** **WISP** random walks; a weight per instance	**now** **8 + 8 + 4 datasets** all rerun, one protocol

Status: implemented and tested; ACM memory 9.3 → 2.8 GB. **No accuracy runs yet.**

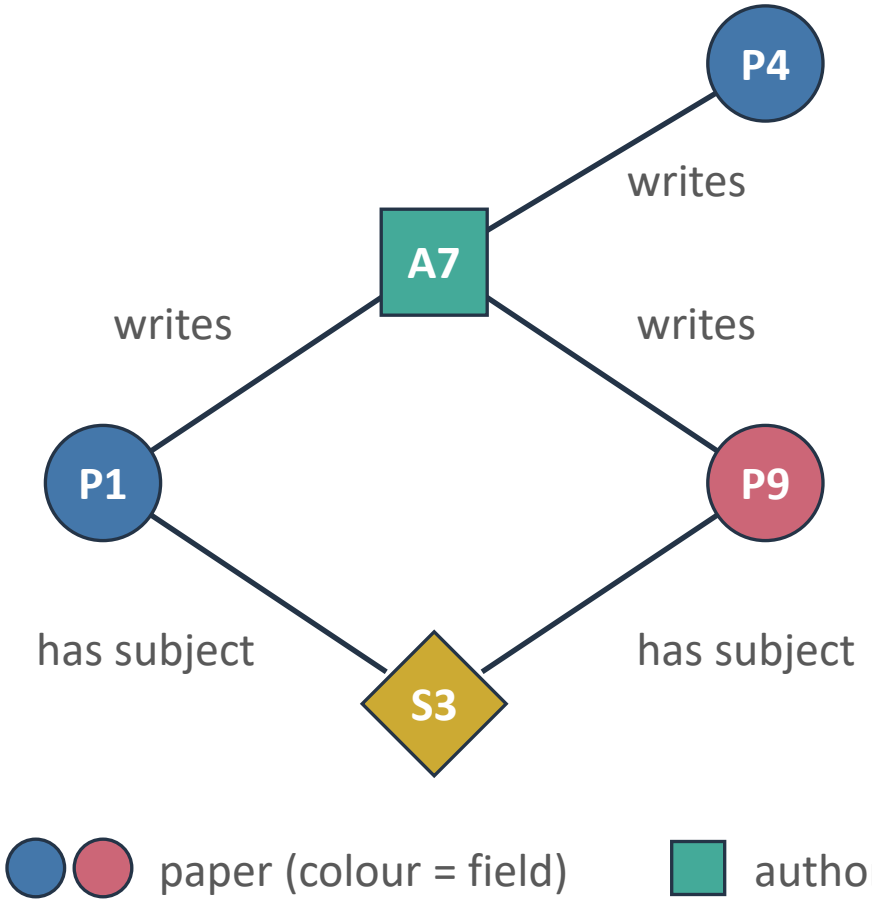
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Outline

1. Background
2. Problem
3. Method
4. Experiments
5. Discussion

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A heterogeneous graph: nodes/edges have types



$$G = (V, E, \phi, \psi), \qquad \phi : V \to \mathcal{T}, \qquad \psi : E \to \mathcal{R}$$

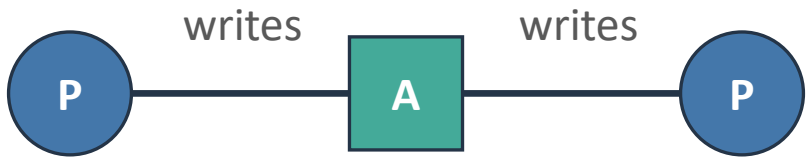
- **Node types** $\mathcal{T}$ = {paper, author, subject}
- **Relation types** $\mathcal{R}$ = {writes, has-subject, cites}
- **Paper (Target) Class**: Math, Biology
- **Example:** Citation networks, Recommendation Systems

Types carry meaning: an author neighbour says something different from a subject neighbour.

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Meta-paths: a type pattern versus one concrete path

Schema (type pattern): PAP

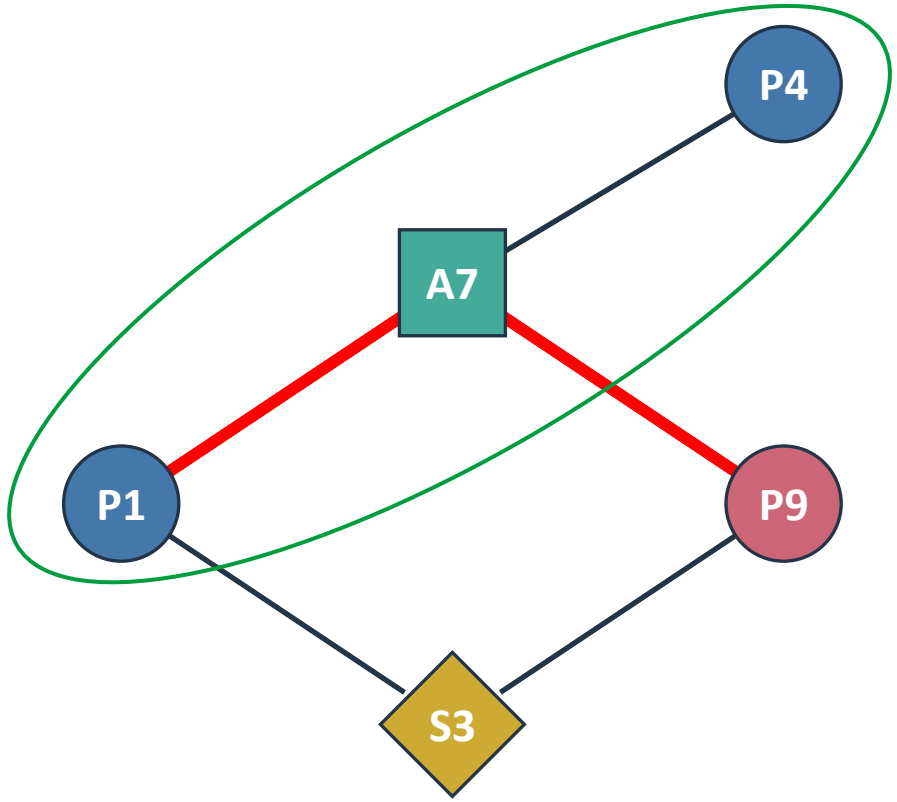


PAP ≈ “two papers share an author”

APA ≈ “Co authors”

Instance (PAP)

- **P1 – A7 – P9**: P1 and P9 share A7
- **P1 – A7 – P4**: P1 and P4 share A7
- A7 is the intermediate node

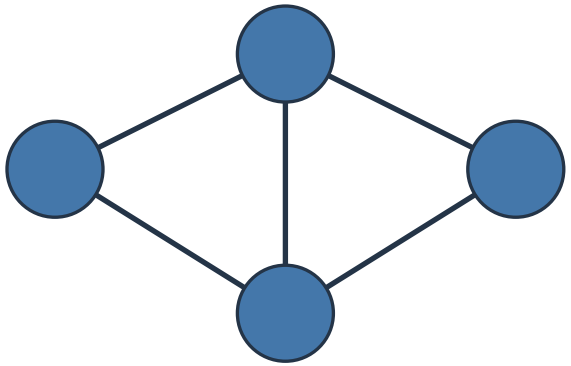


Meta-paths [4] are the standard tool for typed multi-hop relations. It can encode semantic meanings

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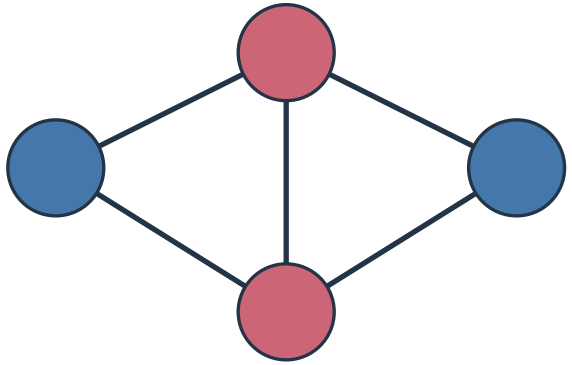
Heterophily: connected nodes have different labels

Homogeneous Homophily



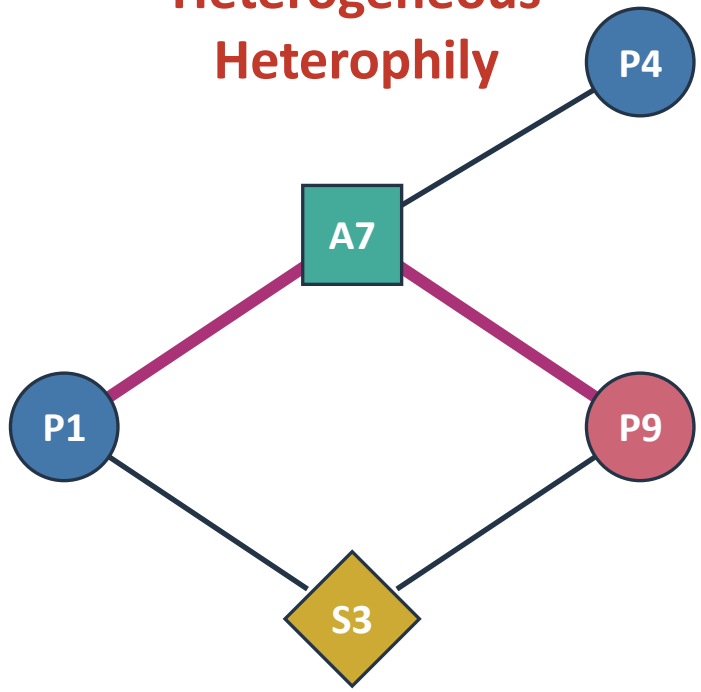
Average the neighbours → label is kept

Homogeneous Heterophily



Average the neighbours → labels are blurred

Heterogeneous Heterophily



P1 and P9 share author A7, but work in different fields

Being connected does not mean being similar; plain averaging (GCN) can hurt.

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Measuring heterophily: H^2[6] on heterogeneous graph

$$\mathcal{H}^2(G) = \frac{1}{|\mathcal{P}_2|} \sum_{p \in \mathcal{P}_2} \left(1 - h_{\text{adj}}(G_p)\right)$$

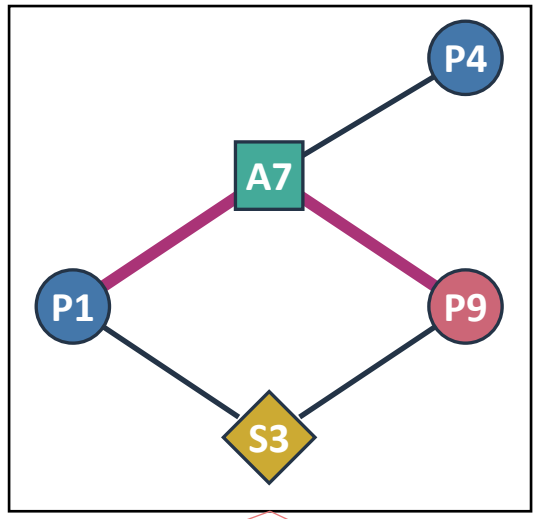
$\mathcal{P}_2$ Homogeneous Subgraph from l =2 meta-paths

G_p paper graph that meta-path p induces

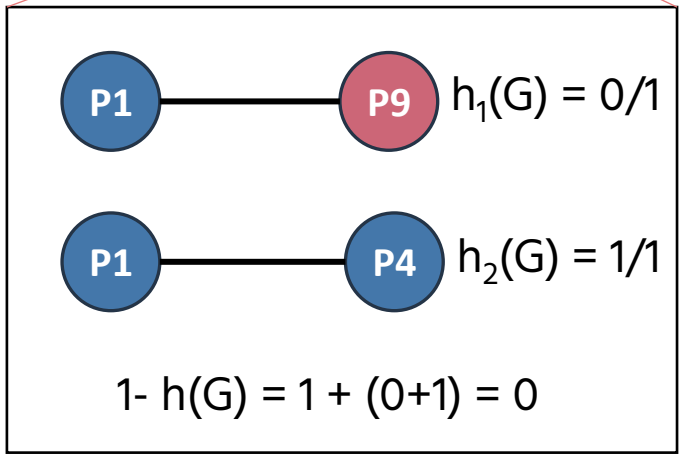
h_{adj} homophily corrected for class imbalance

$$h(G) = \frac{|\{(u,v) \in E : \ y_u = y_v\}|}{|E|}$$

y: node label
near 1: homophily
near 0: heterophily



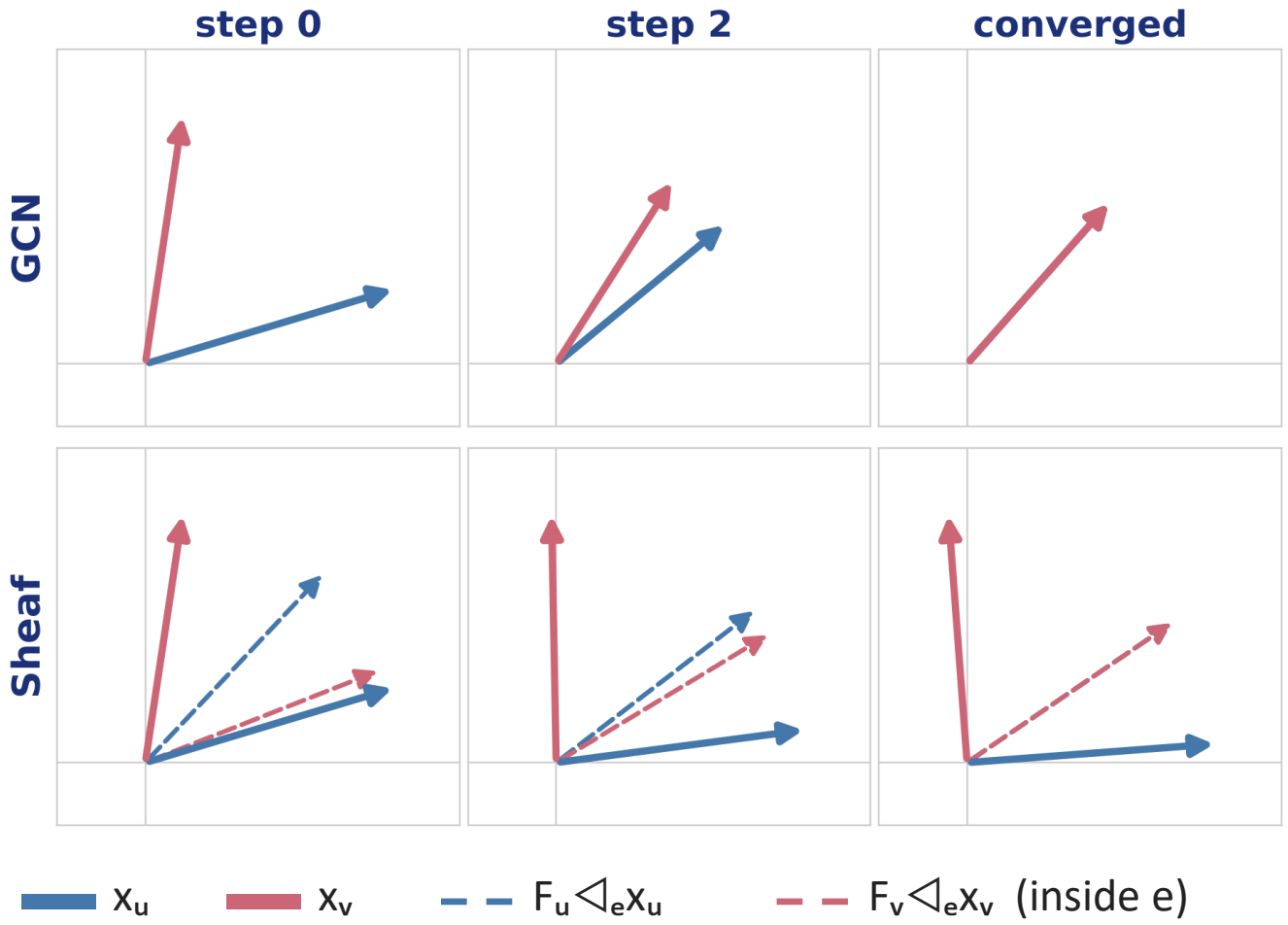
PAP



+ PSP, PPP

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Sheaf diffusion: agree inside the edge, stay different



One step moves both nodes

$$g_e = F_{v \trianglelefteq e} x_v - F_{u \trianglelefteq e} x_u \quad (\text{gap inside } e)$$

$$x_u \leftarrow x_u + \alpha F_{u \trianglelefteq e}^{\top} g_e$$

$$x_v \leftarrow x_v - \alpha F_{v \trianglelefteq e}^{\top} g_e$$

- Both move to close the gap inside e
- GCN: F = I, so the gap is $x_v - x_u$: they merge

Example: F_u rotates +30°, F_v rotates −60°.

Converged: $x_u \approx$ east, $x_v \approx$ north (cleaned, still different); inside e both point the same way.

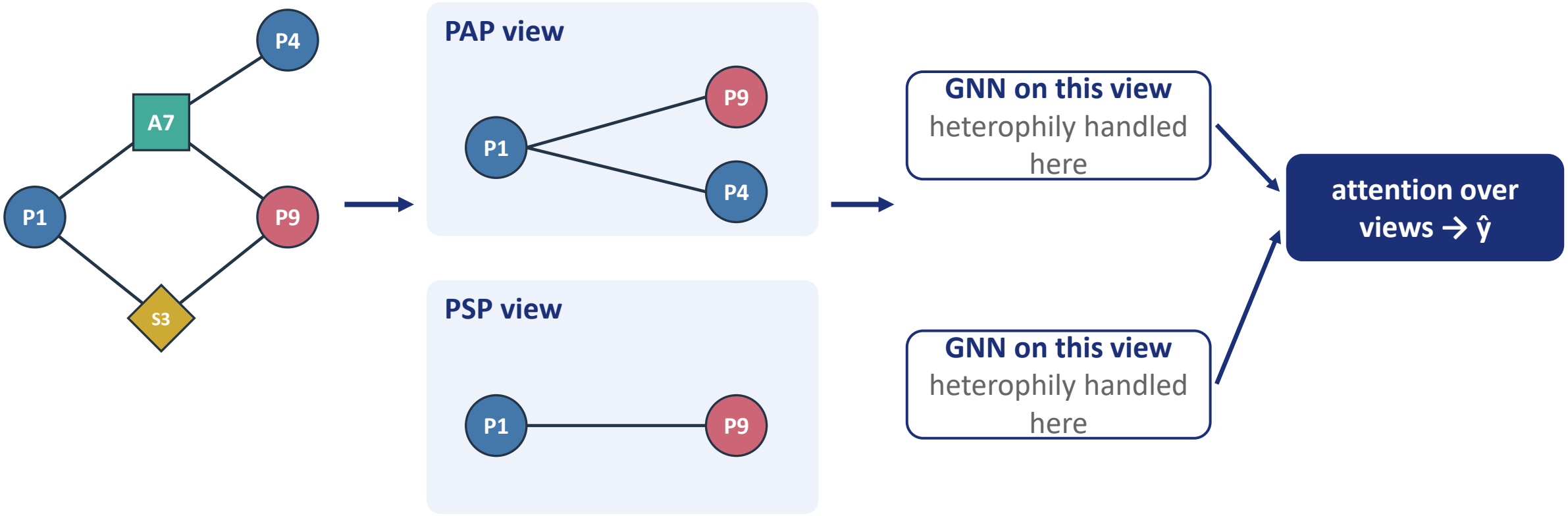
Diffusion removes only the disagreement the maps cannot explain, so each node keeps its class.

GCN [2] · sheaf networks [7, 8] · HetSheaf [9]

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How heterophily is handled: decompose, then fuse

1 Original graph **2 One graph per schema** **3 Encode each view** **4 Fuse**



A7 and S3 are gone

Heterophily is studied indirectly: on derived graphs, one per meta-path schema, then fused.

Decompose-then-fuse [18, 20, 21] · schema-level encoders [10, 11] · per-schema instance encoders [12, 22]

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Two limitations of decompose-then-fuse

Limitation 1 judged on derived graphs

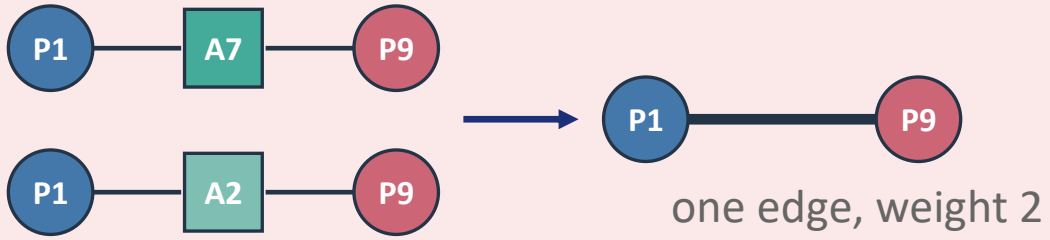
$$h(G_{\mathrm{PAP}}),\quad h(G_{\mathrm{PSP}}),\ \ldots \quad \text{not} \quad h(G)$$

- One homophily per view
- Original typed edges never used
- Depends on a hand-made schema list

Goal: heterophily on the original graph, one model.

Limitation 2 pairs differ only by their endpoints

$$(A_{\mathrm{PAP}})_{P1,P9} = \sum_a (A_{\mathrm{PA}})_{P1,a}(A_{\mathrm{AP}})_{a,P9} = \#\,\text{shared authors}$$



- Only a count survives
- A7 never learns P1–P9 are linked

Goal: path instances act as structure.

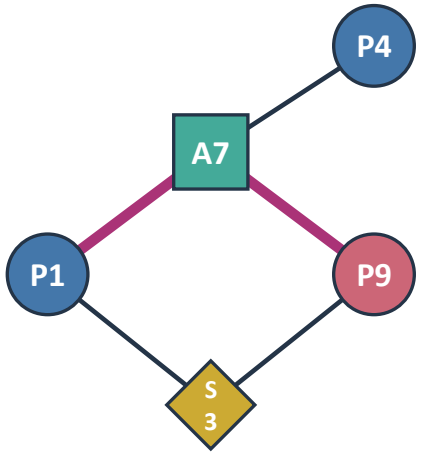
Both limitations come from the same step: the graph is replaced by schema views before learning starts.

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Our view and the research questions

Our view

1. **Heterophily on the original graph**
   no per-schema views
2. **Meta-paths are structure**
   each path instance is a face
3. **Face attached to its nodes and edges**
   no closing edge
4. **A learned weight per instance**
   from its nodes and relations



Running question

Does it matter **who** A7 is? Should A7 learn that P1 and P9 are linked and are part of meta-path?

RQ1 Can a single model on the original graph handle heterophily using path instances, without per-schema branches?

RQ2 Do path instances, used as structure, carry information that schema-level encoders and instance-level features do not?

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Related work: where each family puts the path

Family	Works on the original graph	Uses the path instance	Updates the intermediate node	Where heterophily is handled
Decompose-then-fuse Hetero²Net, HDHGR, LatGRL [18, 20, 21]	✗	✗	✗	per derived view
Schema-level encoders HAN, SeHGNN [10, 11]	✗	✗	✗	not targeted
Instance encoders MAGNN, MECCH [12, 22]	✗	✓ as a feature	✗ target only	not targeted
Sheaf networks NSD, HetSheaf [8, 9]	✓	✗ edges only	n/a	learned edge maps
Cell / hypergraph networks CWN, sheaf hypergraphs [13, 14]	✓	**partly** free maps	✓	not targeted
H2S-NN (ours)	✓	✓ as a face	✓	edge maps + path compositions

No family both stays on the original graph and lets one path update all of its nodes.

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Objectives and contributions

Objectives (what we must show)

O1

One model on the original graph handles heterophily, no per-schema branch (RQ1)

O2

Paths as structure beat schema encoders and instance features (RQ2)

Contributions

1. **Path instances as faces of the graph's sheaf**
   maps composed from edges; reaches every node; no closing edge

2. **One energy, one solve**
   unique for any parameters; no depth to tune

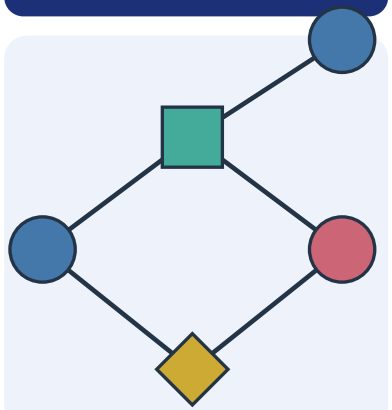
3. **WISP: schema-free sampling**
   random walks, unbiased; a learned weight per instance

Every comparison our questions need is a configuration of the same model on the same sampled paths.

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H2S-NN in one picture

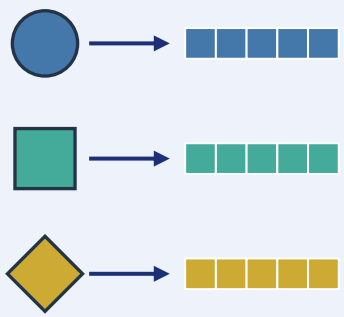
1 Input



$G = (V, E, \phi, \psi)$

types on nodes and edges

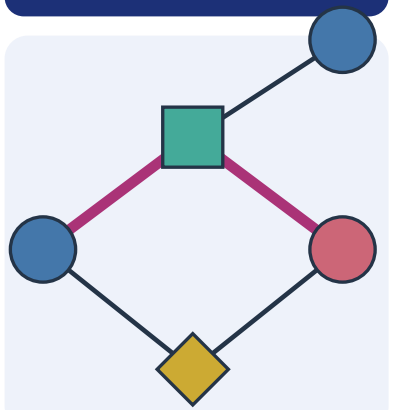
2 Encode



$h_v = W_{\phi(v)} x_v + b_{\phi(v)}$

every type in one d-dim space

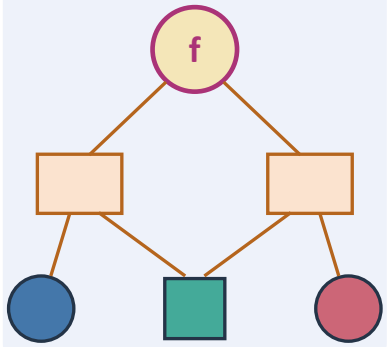
3 Sample faces



$\beta_f = w_f \, m_f$

random walks, a weight each

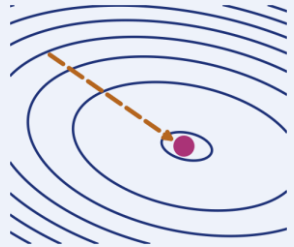
4 Build sheaf



$T_{v,f} = \prod F^\top F$

learned edge maps; face maps composed

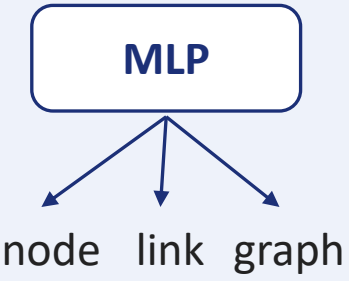
5 Solve



$Y^\star = \arg\min_Y \mathcal{J}(Y)$

agree after maps, stay near h

6 Predict



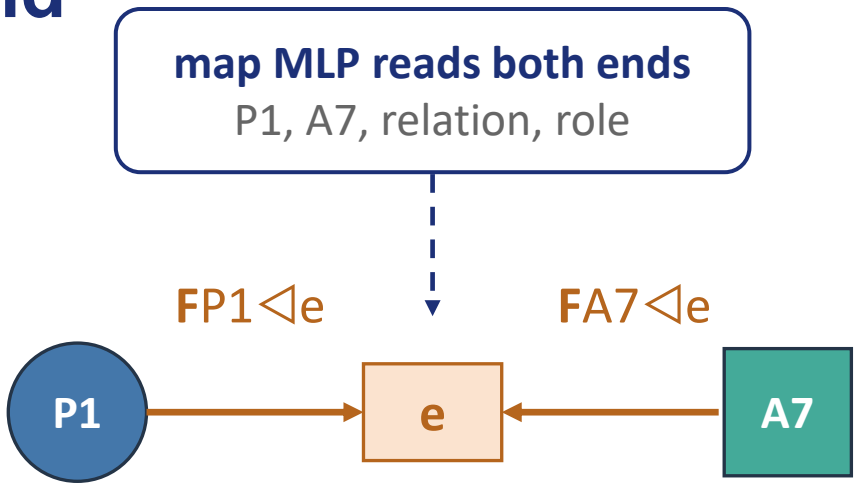
$\hat{y} = \mathrm{MLP}([x^\star, h])$

own + propagated

training gradient: one extra solve, no unrolled layers

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Step 1–2: encode nodes, learn a map for each edge end



- Rotates or reflects, never shrinks
- Depends on who both ends are
- Role frames keep direction

$$h_v = W_{\phi(v)}\, x_v + b_{\phi(v)}$$

type encoder (no features: a learned vector)

$$c_v = \underset{w \in N_r(v)}{\text{mean}}\ h_w$$

context: neighbour mean per relation

$$F_{v \trianglelefteq e} = R_{\psi(e),\text{role}}\, H(u_2)\, H(u_1)$$

edge-end map: frame × two reflections

$$(u_1, u_2) = \text{MLP}\big([h_v, c_v, h_w, c_w, \psi(e), \text{role}]\big)$$

reflections read both ends

$$H(u) = I - 2\,\frac{uu^\top}{\|u\|^2}, \qquad a_e = c_e^2$$

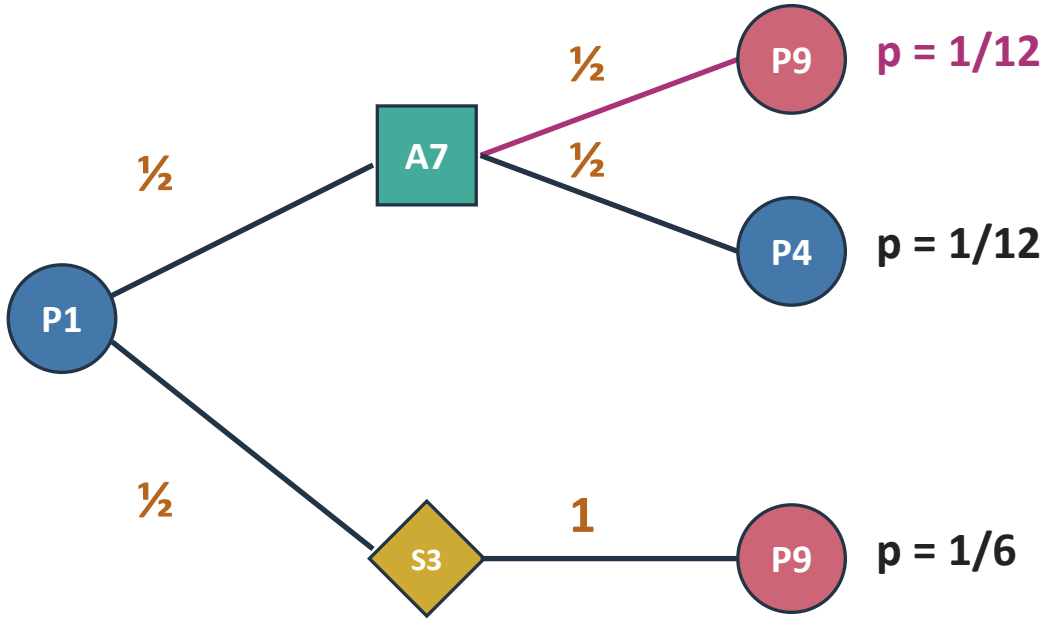
reflection; a_e = edge weight

Computed once per forward pass: the same maps are used by edges and, composed, by faces.

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WISP (1): sample path instances by random walks

All length-2 walks from P1, and their probability



$$p(\mathrm{P1}{-}\mathrm{A7}{-}\mathrm{P9}) = \underbrace{\tfrac{1}{3}}_{\ell=2} \cdot \underbrace{\tfrac{1}{2}}_{\text{action}} \cdot 1 \cdot 1 \cdot \underbrace{\tfrac{1}{2}}_{\text{neighbour}} = \tfrac{1}{12}$$

$$p(\omega) = \frac{1}{\ell_{\max} - 1} \prod_{j=1}^{\ell} \frac{1}{|A_j|} \cdot \frac{1}{|N_j|}$$

uniform length; then uniform action $\mathbf{A_j}$ and unvisited neighbour $\mathbf{N_j}$

$$m_f = \frac{1}{r} \sum_{\text{walks} \to f} \frac{p(\omega)}{\frac{1}{2}p(\omega) + \frac{1}{2}q(\omega)} \leq \frac{2}{r} \text{ per walk}$$

sampling correction; **r** walks per node

- One walk = one face (ℓmax = 4)
- New walks every epoch
- No schema list, no labels

Unbiased: on average the sampled faces give the operator of the stated walk distribution p.

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WISP (2): a learned weight for every instance

Same schema PAP, different weight



w high



w low

Line width = learned weight (illustration)

$$\xi_f = \left[h_{v_0}, \ldots, h_{v_\ell},\ \mathrm{rel}(a_1), \ldots, \mathrm{rel}(a_\ell)\right]$$

all nodes and relations, in order

$$w_f = \exp\left(g_\ell(\xi_f)\right), \qquad \beta_f = w_f\, m_f$$

g: one small MLP per path length

$$I_\pi = \frac{\sum_{f:\,\mathrm{schema}(f)=\pi} \beta_f}{\sum_f \beta_f}$$

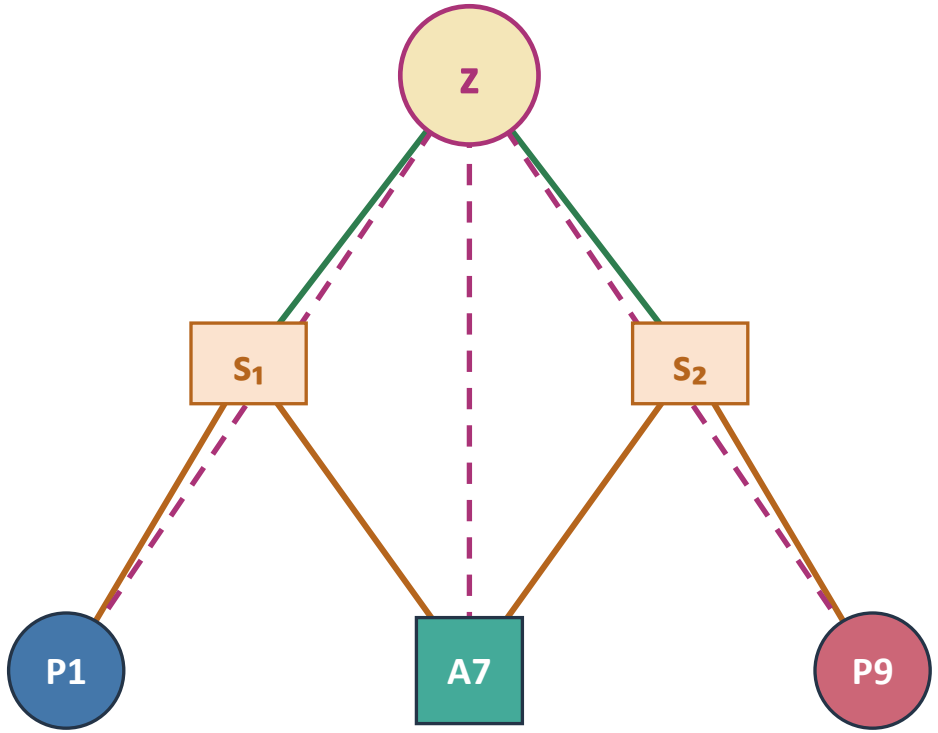
schemas **read out**

v1: schema search + 3 budgets → **WISP:** no schema step

The weight sees who is on the path, so two instances of one schema can count very differently.

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Step 3: a path instance becomes a face



faces (state z)

edges (state s)

nodes (state x)

no P1–P9 edge, not even implicitly

Structure: an order on cells

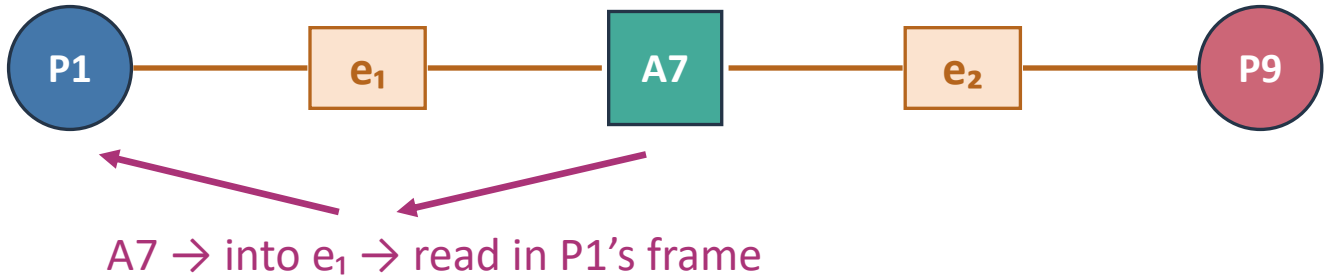
$$\text{node} < \text{edge} < \text{face}$$

- Face sits above its nodes and edges
- Coupled: **node–edge**, **edge–face**, **node–face**
- States x, s, z are solved for
- No loop, no closing edge

A7 is inside the computation: it is coupled to the face directly and through its two edges.

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Face maps are composed from edge maps



$$T_{v_0,f} = I, \qquad T_{v_i,f} = T_{v_{i-1},f}\, F^{\top}_{v_{i-1} \trianglelefteq e_i}\, F_{v_i \trianglelefteq e_i}$$

node → face: walk the path edge by edge

$$T_{A7,f} = F^{\top}_{P1 \trianglelefteq e_1} F_{A7 \trianglelefteq e_1}, \qquad T_{P9,f} = T_{A7,f}\, F^{\top}_{A7 \trianglelefteq e_2} F_{P9 \trianglelefteq e_2}$$

on the example (P1's frame is the face frame)

$$R_{e_i,f} = T_{v_{i-1},f}\, F^{\top}_{v_{i-1} \trianglelefteq e_i}, \qquad R_{e,f}\, F_{v \trianglelefteq e} = T_{v,f}$$

edge → face; every route gives the same map

What this buys

- **No new parameters**
- **Depends on who A7 is**
- **Relation order matters**
- **Never contradicts the graph**
- **Free maps [15] lose this**

A face compares its nodes after carrying them along its own edges.

Step 4: one energy over nodes, edges and faces

$$E(Y) = \frac{1}{2}\sum_{e} a_e \sum_{v \in e} \left\| F_{v \trianglelefteq e} x_v - s_e \right\|^2 + \frac{1}{2}\sum_{f} \beta_f \left[\sum_{e \in f} \left\| R_{e,f} s_e - z_f \right\|^2 + \sum_{v \in f} \left\| T_{v,f} x_v - z_f \right\|^2 \right]$$

node–edge edge–face node–face

Symbol	Meaning	Intuition
Y = (x, s, z)	node, edge, face states, stacked in one vector	what the model solves for
a, β	edge weight (Step 2), face weight (WISP)	how much each agreement counts
F, R, T	maps: F learned (Step 2); R, T composed from F (Step 3)	compare only after the maps
s	meeting point of an edge's ends and its faces	faces sharing an edge must agree there
z	meeting point of all cells of one path	the instance's own state

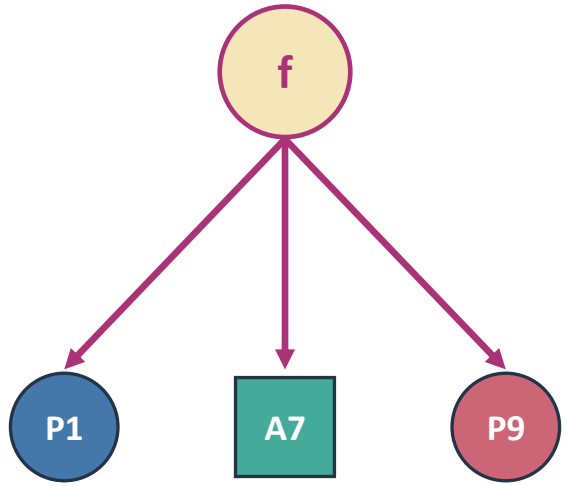
$$E(Y) = \frac{1}{2} Y^\top L Y, \qquad \hat{E}(Y) = \frac{1}{2} Y^\top \hat{L} Y, \quad \hat{L} = D^{-1/2} L D^{-1/2}, \quad D_v = \sum_{e \ni v} a_e + \sum_{f \ni v} \beta_f$$

E is a quadratic in Y with matrix L; Ê is the same energy with degree-normalised weights (hubs do not dominate). Step 5 minimises Ê.

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Every node of the path hears the face

Coupling in the operator	v1: Hodge diffusion + closing edge	H2S-NN
face → P1	0 for any number of steps	direct
face → A7 (intermediate)	0 for any number of steps	direct
face → P9	0 for any number of steps	direct
P1 ↔ P9 shortcut	yes (closing edge)	none



all three receive the face in one application

$$(\hat{L}Y)_v \ \ni \ \beta_f \, T_{v,f}^{\top}\big(T_{v,f}x_v - z_f\big) \qquad \text{for every } v \in f$$

v1: face→node block is exactly 0 (proved); only the closing edge remained.

This removes the cause of reviewer concern 2 by construction; whether it helps accuracy is tested later.

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Step 5: minimise the energy while staying near the

$$\mathcal{J}(Y) = \underbrace{\tfrac{1}{2}\|x-h\|^2}_{\text{stay near the encoding (Step 1)}} + \tau \underbrace{\hat{E}(Y)}_{\text{the Step-4 energy}} \qquad Y^\star = \arg\min_Y \mathcal{J}(Y)$$

Symbol	From
Y = (x, s, z)	all states (Step 4); x = PY is its node part
h	node encodings (Step 1)
Ê, L̂	normalised energy and its matrix (Step 4)
τ = e^θ	one learned scalar (weight of agreement)

Setting the gradient to zero

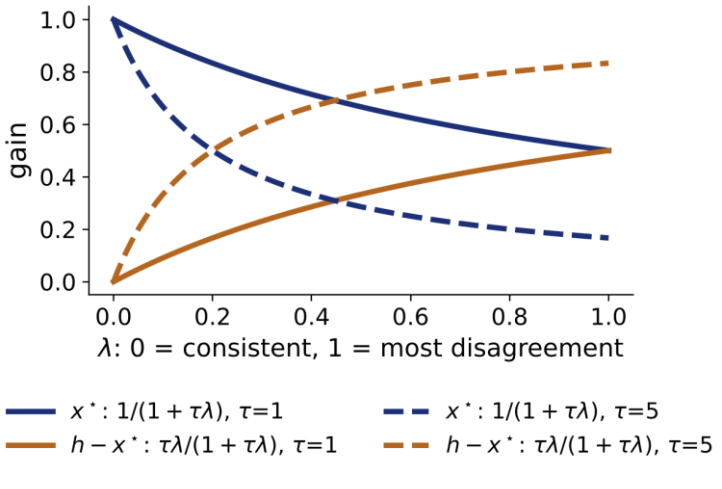
$$\nabla\mathcal{J} = P^\top(PY-h) + \tau\hat{L}Y = 0 \;\Rightarrow\; (P^\top P + \tau\hat{L})\,Y^\star = P^\top h$$

one linear system; conjugate gradient, ≈ 10 steps; unique for any F, a, β, τ

Node view: eliminate s and z exactly

$$x^\star = (I+\tau\hat{S})^{-1}h, \qquad h - x^\star = \tau\hat{S}\,(I+\tau\hat{S})^{-1}h$$

Ŝ: what remains of L̂ on the nodes



x*: smooth part

h – x*: the rest

both reach the head

$$\hat{y} = \mathrm{MLP}([x^\star, h])$$

$\mathcal{J}$ defines the forward pass (no layers, no depth); the training loss comes next and is different.

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What is learned, and what the loss is

Component	Parameters	Enters $\mathcal{J}$ as
Type encoders	W_t, b_t per node type	h
Edge maps	MLP → 2 reflection vectors per edge end; frame per (relation, role)	F
Edge weights	MLP → $a_e > 0$	a_e
Instance weights (WISP)	MLP g per path length → $\beta_f > 0$	β_f
Strength	$\tau = e^{\theta}$, one scalar	τ
Head	MLP on [x*, h]	– (after the solve)

gradient: one adjoint solve with the same matrix; memory ∝ states, not solver steps

Not learned: face maps T (composed from F), the walk sampler, the solver.

$$\min_{\theta} \; \mathcal{L}\big(\mathrm{MLP}_\theta([x^\star, h_\theta]),\, y\big) \quad \text{s.t.} \quad (x^\star, z^\star) = \arg\min \; \mathcal{J}_\theta$$

$\mathcal{L}$: cross-entropy (node, graph), binary (IMDB, links). **$\mathcal{J}$ is not the loss**: it defines the forward pass.

The task loss shapes F, a, β and τ; that is how the model learns what "agree" means on each graph.

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How the method answers the research questions

	Mechanism in the model	Where it is tested
RQ1 heterophily, original graph, no per-schema branch	edge maps F (agree after a map); anchor to h; head reads [x*, h]; one objective on the original graph	gain vs. $\mathcal{H}^2$ across 8 datasets; maps → identity; vs. Hetero^{2}Net, HetSheaf, R-GCN
RQ2 path instances as structure	face state z_f; maps T composed in path order; weight β_f from all nodes of the path; every node on the path updated	no faces; face → closing edge; face → feature; weight uniform / endpoints; known-answer task
Both	WISP: candidates without a schema list; selection per instance, end-to-end	expert list on the same model; LMSPS schemas (proposed); ℓmax curve

Each mechanism was added for one requirement; each has an ablation that can remove it.

Heterophily metrics: labels (MLH) and features (MDE)

MLH: do linked nodes share a label?

$$\mathrm{MLH} = \frac{1}{|\mathcal{M}_2|} \sum_{P} \frac{|\{(u,v) \in E_P : y_u = y_v\}|}{|E_P|}$$

E_P: edges of the graph meta-path P induces; y: label

Purpose: near 1 homophilic, near 0 heterophilic

MDE: do linked nodes look alike?

$$\mathrm{MDE} = \frac{1}{|\mathcal{M}_2|} \sum_{P} \frac{1}{|E_P|} \sum_{(u,v)\in E_P} \frac{1}{4} \left\| \frac{x_u}{\sqrt{|N_u|}} - \frac{x_v}{\sqrt{|N_v|}} \right\|^2$$

x: raw features; N: neighbours in E_P

Purpose: large = feature heterophily; per dataset only

$\mathcal{M}_2$: length-2 meta-paths from Hetero²Net [18] (ACM PAP, PSP, PTP; DBLP APA; IMDB MDM, MAM, MKM), same set for MLH, $\mathcal{H}^2$, MDE. First reproduce Hetero²Net's published values.

Labels and features can disagree; reporting both answers concern 1.

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Datasets (1): node classification, with their

Dataset	Source	Nodes	Target (classes)	MLH ↑het.	$\mathcal{H}^2$ ↑het.	MDE
ACM	HGB [1]	10,942	paper (3)	0.40[H]	0.40	0.74[H]
DBLP	HGB [1]	26,128	author (4)	0.13[H]	0.17	0.80[H]
IMDB	HGB [1]	21,420	movie (5, multi-label)	0.76[H]	0.74	0.59[H]
Freebase	HGB [1]	180,098	book (7)	0.85*	0.995*	–
IEEE-CIS-G	H2GB [5]	153,880	transaction (2)	0.14[G]	0.98[G]	·
mag-year	H2GB [5]	1,939,743	paper (5)	0.77[G]	0.97[G]	·
ogbn-mag	H2GB [5]	1,939,743	paper (349)	0.79[H]	0.88[G]	0.04[H]
H-Pokec	H2GB [5]	1,731,977	user (2)	0.39[G]	0.95[G]	·

Heterophily form (1 = all linked pairs differ). [H] Hetero²Net Table 1, its meta-path set (ACM, DBLP, IMDB reproduced within 0.01) · [G] H2GB Table 2 · unmarked: ours with H2GB's code · * random fill of unlabelled books; labelled-only 0.48 / 0.88 · · to compute

$\mathcal{H}^2$ spans 0.17–0.995. IEEE-CIS-G: MLH looks homophilic, $\mathcal{H}^2$ does not, so $\mathcal{H}^2$ is primary.

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Datasets (2): link prediction and graph classification

Link-prediction task	Graph	Target relation
ACM P–A, DBLP A–P, IMDB M–A	HGB, derived	paper–author, author–paper, movie–actor
Freebase B–B	HGB, derived	book–book (split by unordered pair)
PubMed, LastFM	HGB link sets	disease–disease, user–artist
ogbn-mag P–P, H-Pokec U–U	H2GB graphs	citation, friendship (large)

Leaking features removed in every task.

Graph classification (appendix)	Graphs	Classes	Types
MUTAG, PROTEINS, ENZYMES, NCI1 (TU [23])	188 – 4,110	2 – 6	from atom / structure labels

Graph classification shows coverage only: its types come from labels, so no heterophily claim.

Evaluation metrics and protocol

Macro-F1

$$\frac{1}{|C|}\sum_{c}\frac{2P_cR_c}{P_c+R_c}$$

every class counts

Micro-F1

$$\frac{2PR}{P+R}$$

overall correctness

ROC-AUC

$$\Pr[s(v^+) > s(v^-)]$$

true link above a hard negative

MRR

$$\frac{1}{|Q|}\sum_{q}\frac{1}{\mathrm{rank}_q}$$

true link ranked first

node, graph: Macro/Micro-F1 · link: ROC-AUC, MRR (HGB 2-hop negatives)

Same data files, splits, negatives → **Tune seed 0** 100 Bayesian trials → **Retrain** seeds 0–4 → **Report** mean ± std → **Test** Welch [24], raw + adj. p

Every baseline rerun under this rule, HetSheaf included

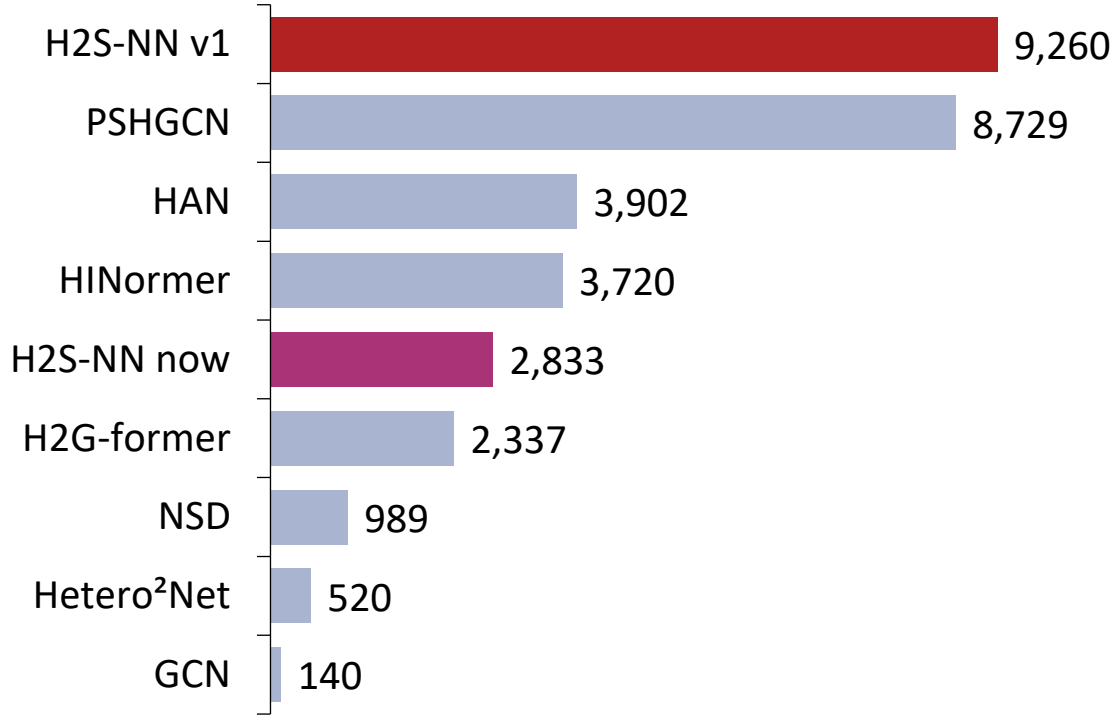
Cost: how it grows, and what we measured

Growth (time unless noted)

Maps	$O(\|E\|\, d)$	no d×d matrices
Faces	$O(I\, d)$	I = face incidences
Solve	$O(T(\|E\| + I)\, d)$	T ≈ 10
Memory	$O((N + \|E\| + F)\, d)$	not ∝ T

v1: O((|E|+I) d^3) maps; **HAN/SeHGNN:** one graph per schema, can be denser than E

Peak GPU memory on ACM (MB)



others: v1 Table 11; now: laptop, d = 32; time on server

Linear in edges and face incidences; ACM memory 9.3 GB → 2.8 GB (concern 3).

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Ablation study: which part carries the gain?

Remove or replace	If the full model wins, it shows	Answers
All faces (β = 0)	path instances add information	RQ2
Face → closing edge	a face is more than its endpoints	RQ2, concern 2
Face → feature of the head	structure beats instance features	RQ2
Weight → uniform / endpoints	who is on the path matters	RQ2
Edge maps → identity	maps are what handle heterophily	RQ1 (proposed)
Face maps → free maps	composing edge maps matters	design

One change per row, same paths and budget; the wording if it fails is fixed in advance.

Free maps as in copresheaf networks [15]; ablations, not rankings: a tuned R-GCN rivals many HGNNs [19]

Model analysis: when and why it works

Gain vs. heterophily

gain plotted against $\mathcal{H}^2$

RQ1

Known-answer tasks

label set by the intermediate node

mechanism

Schema read-out

weight per schema; stable?

interpretability

Path length

ℓmax = 2 … 6

sensitivity

Scaling

memory, time vs. size, #faces

concern 3

Who gets weight

by intermediate type, label match

RQ2

Ablations say which part helps; analyses say when, why, and at what cost.

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What is new, and what is not

New

- Paths as faces of the graph's own sheaf
- Face reaches every node, no shortcut
- Heterophily and meta-paths in one objective
- Ablations as configurations of one model

Not new (we say so)

- Intermediate nodes: MAGNN, SGAT [12, 17]
- Sheaves on typed graphs: HetSheaf [9]
- The solver: personalised PageRank [16]
- A special case equals MAGNN's encoder

The contribution is where the path acts: inside one sheaf on the original graph.

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Then and now: what changed since the submission

Aspect	v1 (submitted)	Now
Story	3 contributions	heterophily on original graph; paths as structure
Face → nodes	only via closing edge	direct, every node
Propagation	K Hodge steps	one objective, one solve
Path selection	AMPSS + 4 budgets	WISP, no schema list
Node datasets	3 (one heterophilic)	8 across heterophily
Baselines	HetSheaf quoted	all rerun, same budget
Memory (ACM)	9.3 GB	2.8 GB

The core idea stayed; its weakest link (the closing edge) and the weak evidence were replaced.

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Limitations and next steps

Limitations (stated in the paper)

- Influence is symmetric in strength; direction lives only in the maps
- Propagation is linear; non-linearity is in the head
- Edge states carry no input of their own (no edge features)

Next steps (in order)

1. Mini-batch design for the large H2GB graphs
2. Baseline plan: coverage, search spaces, HetSheaf adapter
3. Server runs, then the analysis part
4. Analysis & Ablation Studies

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Conclusion

Problem	Heterophily handled on per-schema graphs; paths reduced to endpoints
Idea	Original graph; maps for heterophily; each path instance a face
Method	One objective, one solve; WISP samples and weights paths
Next	RQ1, RQ2 on 8 + 8 + 4 datasets; ablations fixed in advance

Model implemented and tested; no accuracy numbers yet. Thank you.

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Backup slide

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Training: the loss for each downstream task

Node classification

$$\mathcal{L} = -\frac{1}{|S|}\sum_{v\in S}\log \mathrm{softmax}(\hat{y}_v)_{y_v}$$

cross-entropy on labelled training nodes

IMDB (multi-label)

$$\mathcal{L} = -\frac{1}{5|S|}\sum_{v\in S}\sum_{g=1}^{5}\left[y_{vg}\log\sigma(\hat{y}_{vg}) + (1-y_{vg})\log(1-\sigma(\hat{y}_{vg}))\right]$$

one binary loss per genre; predict genre if logit > 0

Link prediction

$$\mathcal{L} = -\sum_{(u,v)\in E^+}\log\sigma(s_{uv}) - \sum_{(u,v)\in E^-}\log\big(1-\sigma(s_{uv})\big), \quad s_{uv} = \mathrm{dec}(z_u, z_v)$$

z = [x*, h]; one negative per positive; decoder as in HGB

Graph classification

$$\hat{y}_G = \mathrm{MLP}\Big(\underset{v\in G}{\mathrm{pool}}\,[x_v^\star, h_v]\Big), \quad \mathcal{L} = \mathrm{CE}(\hat{y}_G, y_G)$$

pool ∈ {mean, max, add, attention}; no new readout

The propagation is the same for all tasks; only the head and the loss change.

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Datasets (1): node classification, small to large

Dataset	Source	Nodes	Node types	Target (classes)	Training
ACM	HGB [1]	10,942	4	paper (3)	full graph
DBLP	HGB [1]	26,128	4	author (4)	full graph
IMDB	HGB [1]	21,420	4	movie (5, multi-label)	full graph
Freebase	HGB [1]	180,098	8	book (7)	full graph*
IEEE-CIS-G	H2GB [5]	153,880	12	transaction (2)	full or mini-batch
mag-year	H2GB [5]	1,939,743	4	paper (5)	mini-batch
ogbn-mag	H2GB [5]	1,939,743	4	paper (349)	mini-batch
H-Pokec	H2GB [5]	1,731,977	16	user (2)	mini-batch

Splits: HGB 24/6/70, H2GB its own. *Freebase memory not yet measured

Four HGB graphs plus four large H2GB graphs, chosen to span the heterophily range.

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Datasets (2): link prediction and graph classification

Link-prediction task	Graph	Target relation
ACM P–A, DBLP A–P, IMDB M–A	HGB, derived	paper–author, author–paper, movie–actor
Freebase B–B	HGB, derived	book–book (split by unordered pair)
PubMed, LastFM	HGB link sets	disease–disease, user–artist
ogbn-mag P–P, H-Pokec U–U	H2GB graphs	citation, friendship (large)

Leaking features removed in every task.

Graph classification (appendix)	Graphs	Classes	Types
MUTAG, PROTEINS, ENZYMES, NCI1 (TU [23])	188 – 4,110	2 – 6	from atom / structure labels

Graph classification shows coverage only: its types come from labels, so no heterophily claim.

Heterophily metric (1): MLH, do linked nodes share a label?

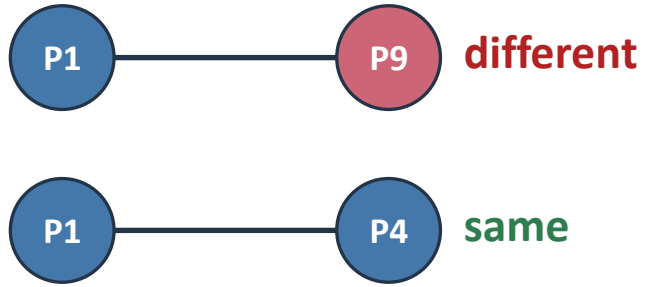
$$\mathrm{MLH}(G) = \frac{1}{|\mathcal{M}_2|} \sum_{P \in \mathcal{M}_2} H_{\mathrm{edge}}(G_P)$$

$$H_{\mathrm{edge}}(G_P) = \frac{|\{(u,v) \in E_P : \; y_u = y_v\}|}{|E_P|}$$

Symbol	Meaning
$\mathcal{M}_2$	meta-paths of length 2, target type to target type
G_P	graph P induces on target nodes (PAP: shared author)
y	class label of a target node
MLH	near 1: linked nodes share labels; near 0: heterophilic

Purpose: label agreement; $\mathcal{H}^2$ [5, 6] corrects it for class imbalance.

Example: PAP on our graph



$$H_{\mathrm{edge}}(G_{\mathrm{PAP}}) = \frac{1}{2}$$

Meta-path set $\mathcal{M}_2$ (from Hetero²Net [18])	
ACM	PAP, PSP, PTP
DBLP	APA
IMDB	MDM, MAM, MKM
ogbn-mag	PFP, PAP
others	its get_metapaths rule

Heterophily metric (2): MDE, do linked nodes look alike?

$$\mathrm{MDE}(G) = \frac{1}{|\mathcal{M}_2|} \sum_{P \in \mathcal{M}_2} \frac{1}{|E_P|} \sum_{(u,v) \in E_P} \frac{1}{4} \left\| \frac{x_u}{\sqrt{|N_u|}} - \frac{x_v}{\sqrt{|N_v|}} \right\|^2$$

Symbol	Meaning
x_u	raw feature vector of target node u
N_u	neighbours of u in G_P (degree normalises)
E_P	edges of the meta-path graph G_P
MDE	large: linked nodes have different features

	Node	Link	Graph
MLH	✓	labelled ends	1-hop*
$\mathcal{H}^2$	✓	labelled ends	–
MDE	✓+	kept features	–

* label = atom type + not Freebase (no features)

Purpose: feature disagreement; describes a dataset (scale-dependent).

Check first: reproduce Hetero²Net [18]: MLH 0.87/0.60/0.24, MDE 0.80/0.74/0.59 (DBLP/ACM/IMDB)

Labels (MLH, $\mathcal{H}^2$) and features (MDE) can disagree; reporting both answers reviewer concern 1.

Evaluation metrics

Macro-F1 (node, graph)

$$\frac{1}{|C|}\sum_{c\in C}\frac{2P_cR_c}{P_c+R_c}$$

Pc, Rc: precision, recall of class c; C: classes in the split

Purpose: every class counts equally, even rare ones

Micro-F1 (node, graph)

$$\frac{2PR}{P+R},\quad P,R \text{ pooled over classes}$$

single-label: equals accuracy

Purpose: overall correctness

ROC-AUC (link)

$$\Pr\left[s(u,v^+) > s(u,v^-)\right]$$

s: model score; v^-: 2-hop negative, 1 per positive

Purpose: true link ranked above a hard negative

MRR (link)

$$\frac{1}{|Q|}\sum_{q\in Q}\frac{1}{\text{rank}_q}$$

rank of the true tail among its head's candidates

Purpose: rewards putting the true link first

Model selection: validation Macro-F1 (node, graph), validation ROC-AUC (link); fixed before running.

Protocol: one rule for every method

Shared data files, splits, negatives → **Tune on seed 0** 100 Bayesian trials → **Retrain** best config, seeds 0–4 → **Report** mean ± std → **Test** Welch, raw + adj. p

$$t = \frac{\bar{a} - \bar{b}}{\sqrt{s_a^2/n_a + s_b^2/n_b}}$$

Welch [24]: unequal variances; vs. the best other method; **a better baseline is reported too**

Reviewer concern	What the protocol does
Baseline tuning not described	same search budget (100 trials) and rule for every method; spaces published
HetSheaf quoted from its paper	every baseline rerun on our data, HetSheaf included
How std was computed	fixed splits; std over 5 training seeds; Welch test on every metric

Each method keeps its own feature preprocessing (as in its code); everything else is shared.

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Hyperparameters: what is tuned, fixed and learned, per step

Step	Tuned (searched)	Fixed (set once, never tuned)	Learned
1 Encode	**d, encoder dropout**	–	type encoders
2 Edge maps	–	2 reflections, MLP width 128	map MLP, frames, a_e
3 WISP	–	ℓmax = 4, walks r (calibrated), q = p	weight MLP g
4 Energy	–	normalisation	–
5 Solve	–	tolerance 1e-4, max 100 steps	τ
6 Head	**classifier dropout**	width 128 (GC: pooling tuned)	MLP
Training	**learning rate, weight decay**	AdamW, cosine, clip 1.0, epochs	–

9 + 4 **v1 (submitted)** tuned + budgets

5 + 2 **Now** tuned + fixed

v1 also tuned diffusion steps K and two step sizes.

Cost (1): how time and memory grow

Step	Time	Memory
WISP draw (CPU)	$O(N\,r\,\ell_{\max})$	$O(F\,\ell_{\max})$
Edge maps	$O(\lvert E\rvert\,k\,d)$	$O(\lvert E\rvert\,k\,d)$
Instance weights g	$O(F\,\ell\,d\,h)$	$O(F\,h)$
Solve: T steps	$O(T\,(\lvert E\rvert + I)\,d)$	$O((N + \lvert E_s\rvert + F)\,d)$
Backward: 1 solve	$O(T\,(\lvert E\rvert + I)\,d)$	independent of T

N nodes, **|E|** edges, **F** faces, **I** face incidences, **k = 2** reflections, **T ≈ 10** CG steps

Compared with

v1

$O\big((M + I)\,d^3\big)$

d×d maps; memory grows with K

HAN, SeHGNN

$O\big(\sum_\pi |E_\pi|\,d\big)$

schema graphs can be denser than E

MAGNN

$O\big(\sum_\pi \#\mathrm{inst}_\pi\,\ell\,d\big)$

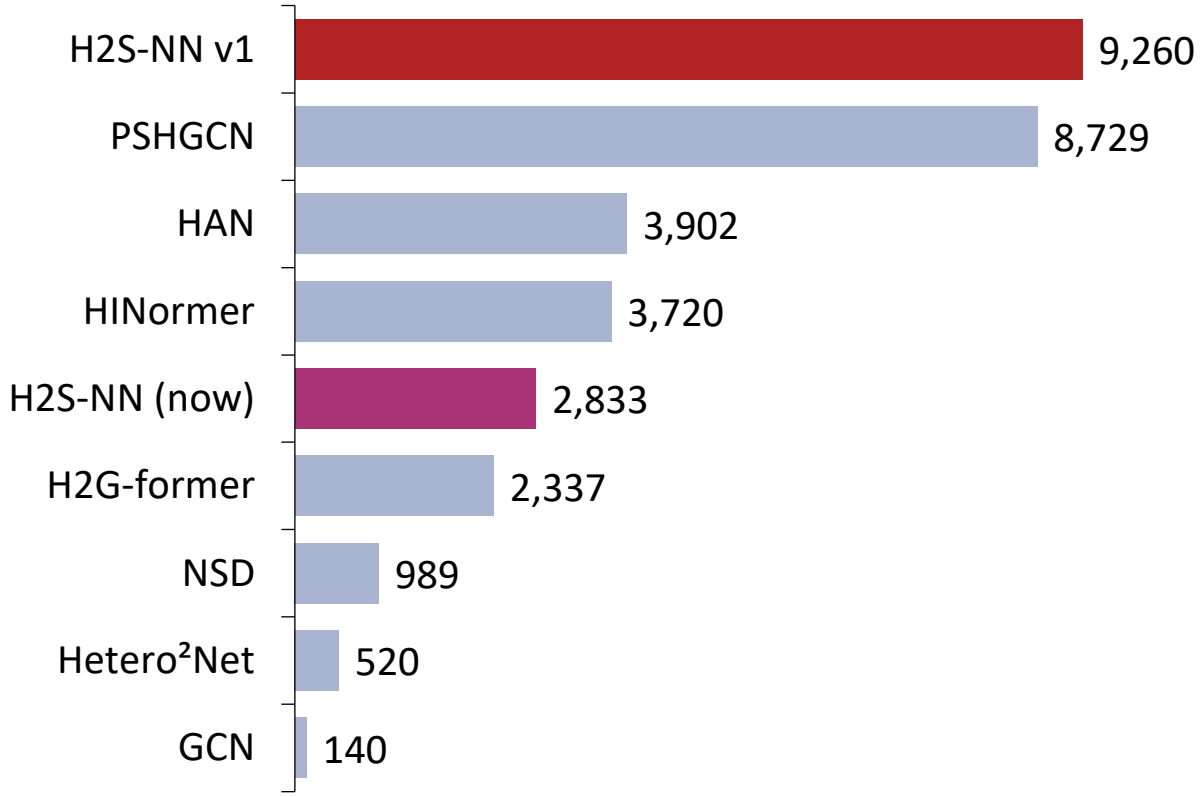
every instance of every schema

Linear in edges and face incidences; d, not d^3; memory does not grow with solver steps.

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Cost (2): measured memory and solver steps

Peak GPU memory on ACM (MB)



H2S-NN now	ACM	DBLP	IMDB
Faces (eval draw)	103k	207k	38k
Peak memory	2.8 GB	5.0 GB	1.3 GB
CG steps (fwd/bwd)	9 / 8	11 / 10	10 / 10
Epoch time	4.3 s	6.2 s	3.1 s

CG steps stay at 9–11 while the state grows 4×.

Time: laptop session ran slow; v1 3.0 s/epoch on ACM; fair timing on the server.

Others: v1 paper Table 11. Now: laptop GPU, d = 32.

Memory on ACM: 9.3 GB (v1) → 2.8 GB now, inside the range of common baselines.

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Ablation study: which part carries the gain?

Remove or replace	If the full model wins, it shows	Answers
All faces (weight 0)	path instances add information	RQ2
Face → closing edge	a face is more than its endpoints	RQ2, concern 2
Face → feature of the head	structure beats instance features	RQ2
Weight: uniform / schema / endpoints	who is on the path matters	thesis, concern 5
Node–face term	a face must reach nodes directly	design
Face maps → free maps	composed maps matter	design
Orthogonal → diagonal maps	relation order matters	design

Same sampled paths, same parameter budget, same tuning; shaded rows decide RQ2.

Each row isolates one component, as reviewers asked; the result we will write is fixed before running.

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Model analysis: when and why it works

Gain vs. heterophily

gain over the best baseline and over “no faces”, plotted against $\mathcal{H}^2$

RQ1, concern 1

Known-answer tasks

synthetic graphs where the label depends on the intermediate node

mechanism

Schema read-out

share of weight per schema; stable across seeds?

interpretability

Path length

accuracy for ℓmax = 2 … 6

sensitivity

Scaling

memory, time, CG steps vs. graph size and #faces

concern 3

Who gets weight

weights of instances by intermediate type and label agreement

RQ2, case study

Ablations say which part helps; analyses say when, why, and at what cost.

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What is new, and what is not

New

- Paths as faces of the graph's own sheaf
- Face reaches every node, no shortcut
- Heterophily on the original graph
- Controls are configurations of one model

Not new (we say so)

- Intermediate nodes: MAGNN, SGAT [12, 17]
- Sheaves on typed graphs: HetSheaf [9]
- The solver: personalised PageRank [16]
- Special case = MAGNN's instance encoder

The contribution is where the path acts (inside one sheaf on the original graph), not that paths are used.

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Then and now (1/2): story and model

Aspect	v1 (submitted, WSDM 2027)	Now
Story	keep path instances as 2-cells; 3 contributions	heterophily on the original graph; paths as structure; 2 goals
Path cell	2-cell closed by a synthetic closing edge	face above its path; no closing edge
Face → nodes	only through the closing edge (proved 0 otherwise)	direct, to every node of the path
Propagation	K Hodge diffusion steps on nodes and edges	one energy, one solve; no depth K
Path selection	AMPSS: DFS over schemas + 3 budgets	WISP: random walks, no schema list
Instance weight	within selected schemas	every instance, from its nodes and relations
Graph readout	SheafCovPool (new component)	plain pooling; no new component
Memory, ACM	9.3 GB	0.30× v1 (laptop GPU, d = 32)

Two components were dropped; the core idea was kept and its weakest link (the closing edge) removed.

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Then and now (2/2): evaluation

Aspect	v1 (submitted)	Now
Node classification	DBLP, ACM, IMDB (one strongly heterophilic)	8 datasets incl. 4 large H2GB graphs
Link prediction	Amazon, YouTube (one node type), LastFM, PubMed	8 tasks with real types; leak-free features
Graph classification	10-fold × 5 seeds; claimed heterogeneous	HetSheaf's protocol; typed by label, stated
Heterophily evidence	MLH only	MLH, $\mathcal{H}^2$, MDE; gains vs. $\mathcal{H}^2$
Baselines	HetSheaf quoted; tuning not described	all rerun, 100 Bayesian trials each
Statistics	mean ± std	5 seeds + Welch's t-test, raw and adjusted p
Ablations	single dataset, within std	controls as configurations, pre-registered

Each row answers one reviewer concern; none of it has been run yet.

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Limitations and next steps

Limitations (stated in the paper)

- Influence is symmetric in strength; direction lives only in the maps
- Propagation is linear; non-linearity is in the head
- Edge states carry no input of their own (no edge features)

Next steps (in order)

1. Mini-batch design for the large H2GB graphs
2. Baseline plan: coverage, search spaces, HetSheaf adapter
3. Server runs, then the analysis part
4. Analysis & Ablation Studies

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Conclusion

Problem	Heterophily handled on per-schema graphs; paths reduced to endpoints
Idea	Original graph; each path instance a face reaching all its nodes
Method	One energy, one solve; WISP samples and weights paths
Evidence plan	RQ1, RQ2 on 8 + 8 + 4 datasets; controls fixed in advance

Status: model implemented and checked; no accuracy numbers yet. Thank you.

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Appendix: terms used in this talk

Term	Meaning
Heterogeneous graph	nodes and edges have types
Heterophily	linked nodes tend to have different labels (or features)
Meta-path schema	a type pattern, e.g. paper–author–paper (PAP)
Path instance	one concrete walk that follows it, e.g. P1–A7–P9
Restriction map F	learned map from a node into the space of an edge
Face	a cell placed above one sampled path instance (no closing edge)
Energy	sum of disagreements between touching cells, after the maps
WISP	Walk-sampled, Instance-weighted Sheaf Paths: the sampler + weights
τ	learned strength of agreement in the solve